Q1

Student ID → xx ... 4 3 2 1

last four digits

$$A = \left[(2+1) + 1 \right] = 2$$

fixed

$$A = \boxed{?}$$

Write only the answer, no calculations!!!

$$1 \leq A \leq 19$$

Q2

calculated in Q1

$V_A = A + 3$ V

7.27 An NMOS transistor has $\mu_n C_{ox} = 400\ \mu A/V^2$, $W/L = 20$, $V_t = 0.5$ V, and ~~$V_A = 5$ V~~. Find g_m and r_o when (a) the bias voltage $V_{GS} = 0.75$ V, (b) the bias current $I_D = 0.5$ mA.

a) $g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) = 400\mu \cdot 20 \cdot (0.25) = 2\,mA/V$

$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2 = 0.25\,mA$

$r_o = \frac{V_A}{I_D} = \frac{A+3}{0.25m} = (4A+12)\,k\Omega$

b) $g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} \cdot I_D} = \sqrt{2 \cdot 400\mu \cdot 20 \cdot \frac{1}{2} m} = 2.83\,mA/V$

$r_o = \frac{V_A}{I_D} = \frac{A+3}{0.5m} = (2A+6)\,k\Omega$

Q3

$L = A\ \mu m$

7.29 An NMOS technology has $\mu_n C_{ox} = 400\ \mu A/V^2$, $V_t = 0.5$ V, and $V_A' = 6V/\mu m$. For a transistor with ~~$L = 0.5$~~ μm, find r_o and the value of W that results in $g_m = 2$ mA/V at $I_D = 0.25$ mA. Also, find r_o and the required V_{GS}.

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = 2m\ A/V = \sqrt{2 \cdot 400\mu \cdot (W/L) \cdot 1/4m} \Rightarrow W/L = 20$$

$$W = \frac{W}{L} \cdot L = 20 \cdot A\mu = 20A\ \mu m$$

$$r_o = \frac{V_A' \cdot L}{I_D} = \frac{6V/\mu m \cdot A\mu m}{0.25m} = 24k\Omega$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2 = \frac{1}{4}m = \frac{1}{2} \cdot 400\mu \cdot 20\ V_{ov}^2$$

$$V_{ov} = 1/4V = V_{GS} - V_t = 0.25V$$

$$V_{GS} = 0.25 + 0.5 = 0.75V$$

Q4

7.30 For the NMOS amplifier in Fig. P7.30, replace the transistor with its T equivalent circuit, assuming $\lambda = 0$. Derive expressions for the voltage gains v_s/v_i and v_d/v_i.

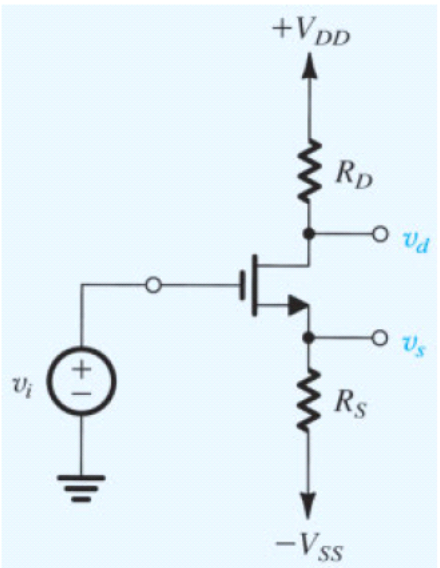
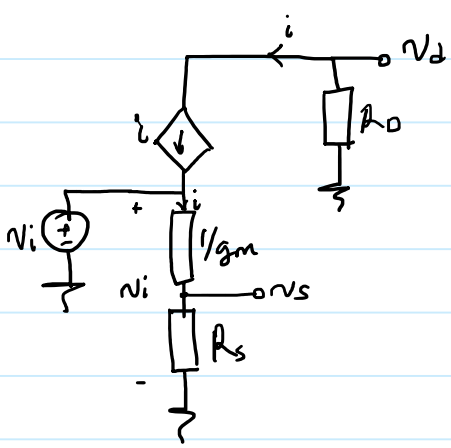


Figure P7.30



$$v_s = \frac{v_i \cdot R_s}{1/g_m + R_s}$$

$$\boxed{v_s/v_i = \frac{g_m R_s}{1 + g_m R_s}}$$

$$v_d = -iR_D,\quad i = \frac{v_i}{1/g_m + R_s} \Rightarrow v_d = -\frac{g_m v_i}{1+g_m R_s} R_D \Rightarrow \boxed{\frac{v_d}{v_i} = -\frac{g_m R_D}{1+g_m R_s}}$$

Q5

7.33 Figure P7.33 shows a discrete-circuit amplifier. The input signal v_{sig} is coupled to the gate through a very large capacitor (shown as infinite). The transistor source is connected to ground at signal frequencies via a very large capacitor (shown as infinite). The output voltage signal that develops at the drain is coupled to a load resistance via a very large capacitor (shown as infinite). All capacitors behave as short circuits for signals and as open circuits for dc.

(a) If the transistor has $V_t = 1$ V, and $k_n = 4$ mA/V^2, verify that the bias circuit establishes $V_{GS} = 1.5$ V, $I_D = 0.5$ mA, and $V_D = +7.0$ V. That is, assume these values, and verify that they are consistent with the values of the circuit components and the device parameters.

(b) Find g_m and r_o if ~~$V_A = 100$ V~~. $V_A = (5A + 50)$ V

(c) Draw a complete small-signal equivalent circuit for the amplifier, assuming all capacitors behave as short circuits at signal frequencies.

(d) Find R_{in}, v_{gs}/v_{sig}, v_o/v_{gs}, and v_o/v_{sig}.

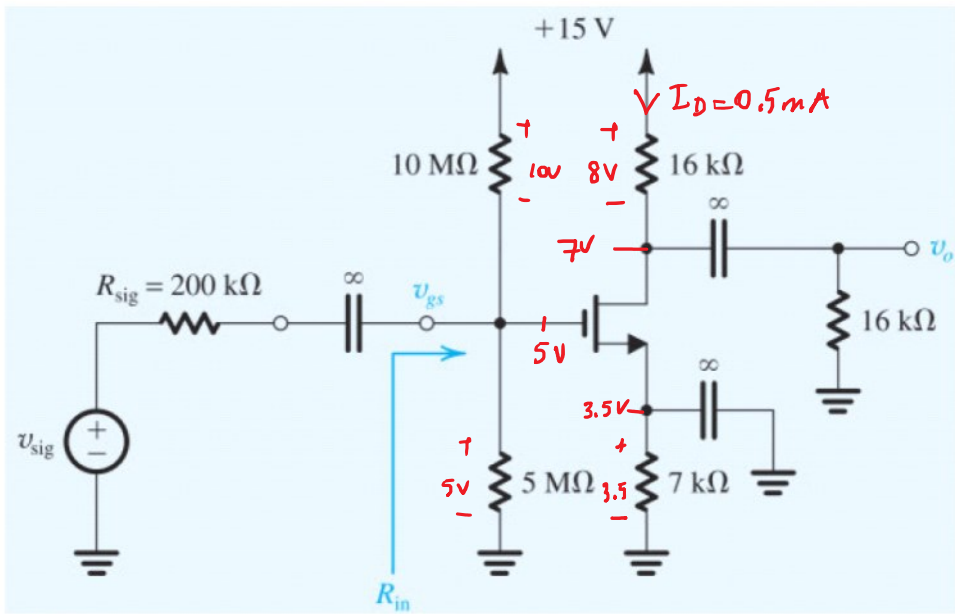


Figure P7.33

$V_{GS} = 1.5$

$V_{ov} = V_{GS} - V_t = 0.5V$

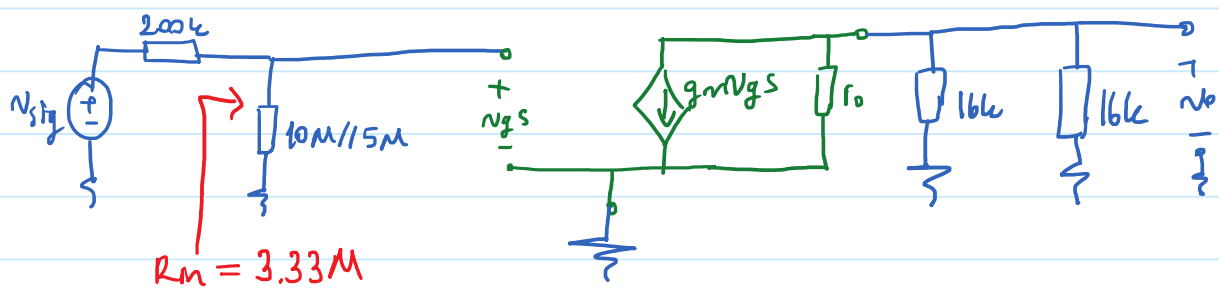
$I_D = \frac{1}{2} k_n V_{ov}^2$

$\frac{1}{2}m = \frac{1}{2} \cdot 4m (\frac{1}{2})^2$ ✓

$g_m = \frac{2I_D}{V_{ov}} = \frac{2 \cdot \frac{1}{2}m}{1/2}$

$g_m = 2 mA/V$

$r_o = \frac{V_A}{I_D} = \frac{[10A+100]k\Omega}{}$



$\frac{v_{gs}}{v_{sig}} = \frac{3333k}{200k + 3333k} = 0.944$

$v_o = -g_m v_{gs} (r_o // 16k // 16k)$

$\frac{v_o}{v_{gs}} = -2m \left[(10A+100) // 8 \right] k$

$\frac{v_o}{v_{sig}} = \frac{v_o}{v_{gs}} \frac{v_{gs}}{v_{sig}} = -0.944 \left(\frac{80A+800}{5A+54} \right)$ ⇐ $\frac{v_o}{v_{gs}} = -2 \cdot \frac{80A+800}{10A+108} = -\frac{80A+800}{5A+54}$

Q6

$\beta = 100 + 5A$

7.36 A transistor with ~~$\beta = 100$~~ is biased to operate at a dc collector current of 0.4 mA. Find the values of g_m, r_π, and r_e. Repeat for a bias current of 40 μA.

@ $I_C = 0.4m$ $\quad g_m = \frac{0.4m}{0.025} = 16mA/V$ $\quad r_e = \frac{\alpha}{g_m} \cong 62\Omega$ $\quad r_\pi = \frac{\beta}{g_m} = \frac{100+5A}{0.016}$

@ $I_C = 0.04mA$ $\quad g_m = \frac{0.04}{0.025} = 1.6mA/V$ $\quad r_e = \frac{\alpha}{g_m} = 618\Omega$ $\quad r_\pi = \beta/g_m = \frac{100+5A}{0.0016}$

Q7

$(4000 + 10A)\,\Omega$

D 7.38 A designer wants to create a BJT amplifier with a g_m of 20 mA/V and a base input resistance of ~~4000 Ω~~ or more. What collector-bias current should she choose? What is the minimum β she can tolerate for the transistor used?

$$g_m = \frac{I_C}{V_T} = 20\,mA/V \Rightarrow I_C = 0.5\,mA$$

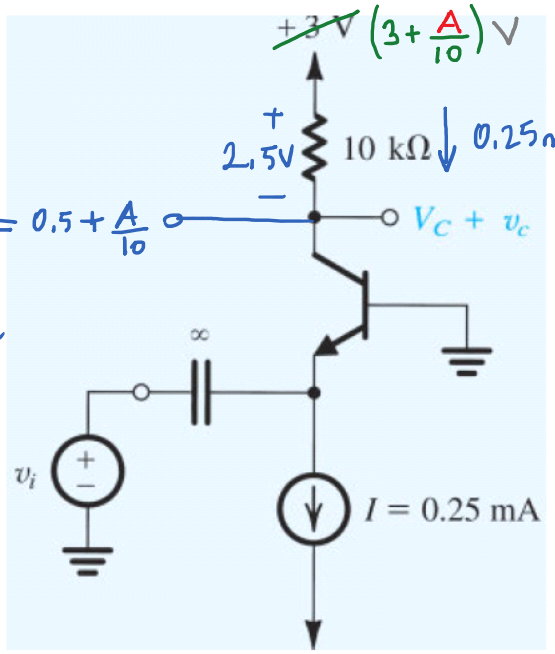
$$r_\pi \geq 4000 + 10A \leq \frac{\beta}{g_m} = \beta/_{0.02} = 50\beta$$

$$\beta_{min} = \frac{4000 + 10A}{50} = 80 + \frac{A}{5}$$

Q8

7.48 The transistor amplifier in Fig. P7.48 is biased with a current source I and has a very high β. Find the dc voltage at the collector, V_C. Also, find the value of r_e. Replace the transistor with the T model of Fig. 7.27(b) (note that the dc current source I should be replaced with an open circuit). Hence find the voltage gain v_c/v_i.

$\beta \to \infty$, $\alpha \to 1$



$r_e = \frac{V_T}{I_E} = \frac{25m}{0.25m} = 100\Omega$

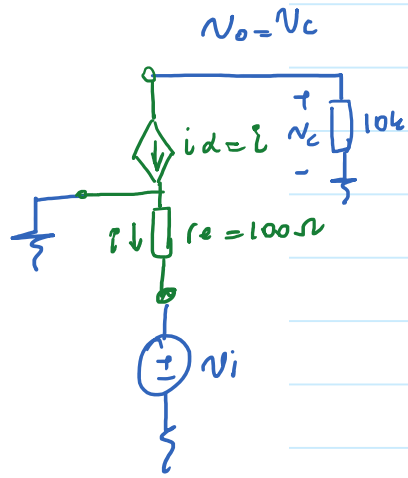
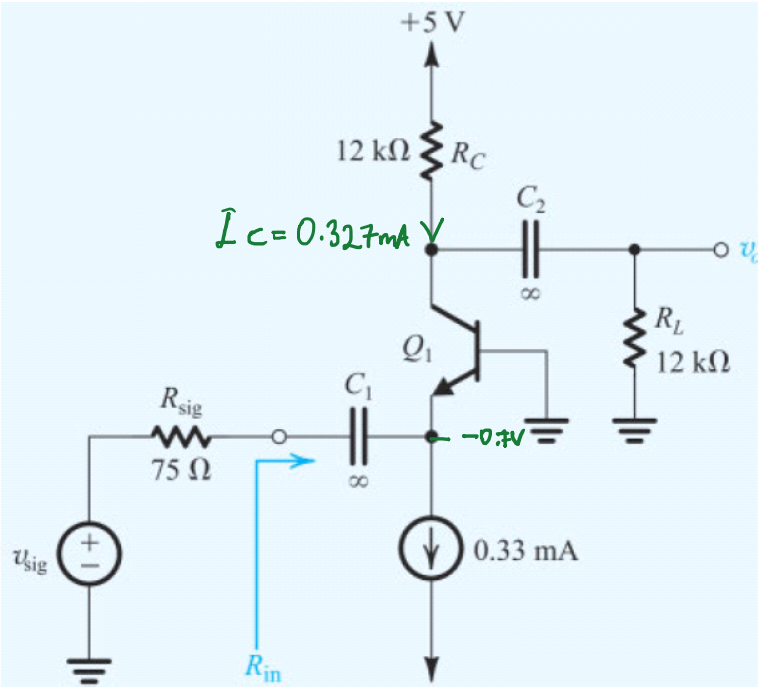


Figure P7.48

$i = -v_i/100 \qquad v_o = -10k \cdot i = 10k\, v_i/100 \Rightarrow \boxed{v_o/v_i = 100}$

Q9

7.53 For the circuit shown in Fig. P7.53, draw a complete small-signal equivalent circuit using an appropriate T model for the BJT (use $\alpha = 0.99$). You should show the values of all components, including the model parameters. What is the input resistance R_{in}? Calculate the overall voltage gain (v_o/v_{sig}).



$g_m = \frac{I_C}{V_T} = \frac{0.327m}{0.025}$

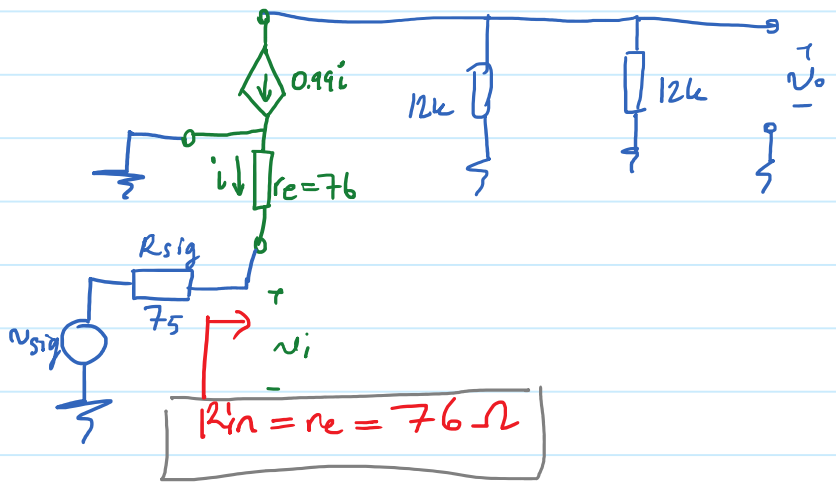
$g_m \cong 13.1$ mA/V

$r_e = \frac{V_T}{I_E} = \frac{25m}{0.33m}$

$= 76\ \Omega$

$\beta = \frac{\alpha}{1-\alpha} = 100$

Figure P7.53



$\frac{v_i}{v_{sig}} = \frac{76}{75+76} \cong 1/2$

$i = -v_i/76$

$v_o = -0.99 i \cdot 6k = 0.99\, v_i \frac{6000}{76}$

$\frac{v_o}{v_i} \cong 78$

$\frac{v_o}{v_{sig}} = \frac{v_o}{v_i} \cdot \frac{v_i}{v_{sig}} \cong 39$ V/V

Q10

(200 + 5·A)

7.54 In the circuit shown in Fig. P7.54, the transistor has a β of ~~200~~. What is the dc voltage at the collector? Replacing the BJT with one of the hybrid-π models (neglecting r_o), draw the equivalent circuit of the amplifier. Find the input resistances R_{ib} and R_{in} and the overall voltage gain (v_o/v_{sig}). For an output signal of ±0.3 V, what values of v_{sig} and v_b are required?

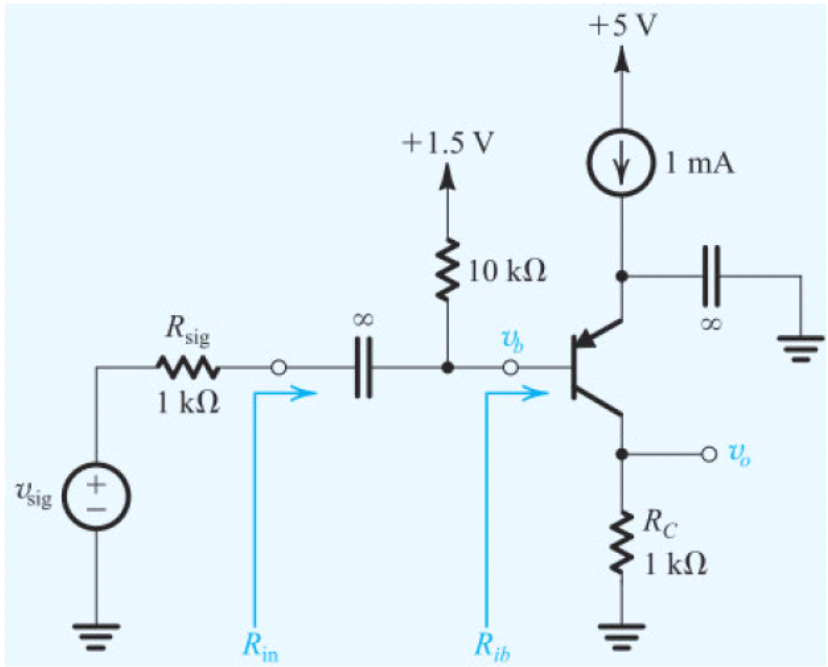


Figure P7.54

$\beta = 200 + 5.A$

$I_E = 1\,mA$

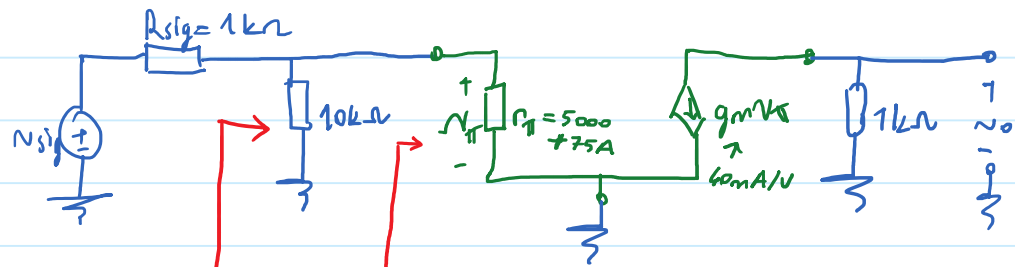
$I_B = 1mA / [200 + 5A]$

$V_B = 1.5 + 10k \cdot I_B$

$g_m = \frac{I_C}{V_T} = \frac{1m}{0.025} = 40\frac{mA}{V}$

$r_\pi = \frac{V_T}{I_B} = \frac{25m}{1m} \cdot [200 + 5A]$

$r_\pi = 5000 + 75A$



$R_{ib} = r_\pi = [5000 + 125.A]\,\Omega$

$R_{in} = R_{ib} // 10k\Omega$

$$G_V = \frac{v_o}{v_{sig}} = \frac{v_o}{v_\pi} \cdot \frac{v_\pi}{v_{sig}} = -g_m(1k) \cdot \frac{R_{in}}{R_{in} + 1k}$$

$$v_o = 0.3V \xrightarrow{G_V} v_{sig} = \frac{v_o}{G_V} = \frac{0.3}{G_V}$$